PREPRINT · KOUROVKA NOTEBOOK 20.87

Finite simple groups whose element orders have at most two prime divisors

A solution of Problem 20.87 of the Kourovka Notebook

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Abstract. We answer Problem 20.87 of the Kourovka Notebook (W. J. Shi): the non-abelian composition factors of finite groups in which every element order is divisible by at most two primes are exactly the finite non-abelian simple groups with this property, and we classify them. Besides the alternating groups A_n, $5 \le n \le 11$, and thirteen sporadic groups, they are the groups $L_2(q)$, $L_3(q)$, $U_3(q)$, $Sz(q)$, ${}^2G_2(q)$ satisfying explicit arithmetic conditions on q, and eighteen further groups of Lie type.

AI disclosure. This solution was obtained by an AI system (AutoPasha, an AI employee, autopasha.ru) and checked by independent computer verification (GAP 4.14 and its Character Table Library). It has not yet been reviewed by a human mathematician; the GAP scripts are available from the authors.

Problem 20.87. *What are the non-abelian composition factors of finite groups in which the order of every element is divisible by at most two primes?* (For one prime, i.e. EPPO-groups, see arXiv:2003.09445.)

Throughout, $\omega(n)$ is the number of distinct prime divisors of a positive integer n ($\omega(1) = 0$), and we say that a finite group X has **property (P2)** if $\omega(|x|) \le 2$ for every $x \in X$, i.e. no element order of X is divisible by three distinct primes. p always denotes the defining characteristic of a group of Lie type over $\mathbb{F}_q$.

0. The answer

Main Theorem. A finite non-abelian simple group T is a composition factor of a finite group with property (P2) if and only if T itself has (P2), and this happens exactly when T is isomorphic to one of the following groups.

1. **Alternating groups:** A_n, $5 \le n \le 11$.
2. **Sporadic groups:** $M_{11}, M_{12}, M_{22}, M_{23}, M_{24}, J_1, J_2, J_3, HS, He, Ru, Suz, O'N$.
3. $L_2(q)$, $q \ge 4$, with $\omega(\frac{q-1}{d}) \le 2$ and $\omega(\frac{q+1}{d}) \le 2$, where $d = (2, q-1)$.
4. $L_3(q)$ with $\omega(\frac{q-1}{d}) \le 1$, $\omega(\frac{q^2-1}{d}) \le 2$, $\omega(\frac{q^2+q+1}{d}) \le 2$, where $d = (3, q-1)$.
5. $U_3(q)$, $q \ge 3$, with $\omega(\frac{q+1}{d}) \le 1$, $\omega(\frac{q^2-1}{d}) \le 2$, $\omega(\frac{q^2-q+1}{d}) \le 2$, where $d = (3, q+1)$.
6. $Sz(q) = {}^2B_2(q)$, $q = 2^{2m+1} \ge 8$, with $\omega(q-1) \le 2$ and $\omega(q \pm \sqrt{2q} + 1) \le 2$.
7. ${}^2G_2(q)$, $q = 3^{2m+1} \ge 27$, with $\omega(q-1) \le 2$, $\omega(q+1) \le 2$ and $\omega(q \pm \sqrt{3q} + 1) \le 2$.
8. **Eighteen further groups of Lie type:** $L_4(3)$, $L_5(2)$, $U_4(2) \cong PSp_4(3)$, $U_4(3)$, $U_5(2)$, $U_6(2)$, $PSp_4(4)$, $PSp_4(8)$, $PSp_6(2)$, $\Omega_7(3)$, $P\Omega_8^+(2)$, $P\Omega_8^+(3)$, $G_2(3)$, $G_2(4)$, $G_2(8)$, ${}^3D_4(2)$, ${}^2F_4(2)'$, ${}^2F_4(8)$.

(Isomorphisms such as $A_5 \cong L_2(4) \cong L_2(5)$, $A_6 \cong L_2(9)$, $A_8 \cong L_4(2)$, $L_2(7) \cong L_3(2)$, $U_3(3) \cong G_2(2)'$, $L_2(8) \cong {}^2G_2(3)'$ cause harmless repetitions.)

All groups of Lie type not covered by items 3–8 fail (P2); in particular every classical group whose natural module has dimension ≥ 9, every group of Lie type of BN-rank ≥ 2 over a field with $q \notin \{2,3,4,8\}$ other than $L_3(q)$, and every group of type $F_4, E_6, {}^2E_6, E_7, E_8$ fails (P2).

Members of the families with small q (computed from the criteria and cross-checked by an independent method, §10):

- $L_2(q)$, $q \leq 1000$ (86 values in $4 \leq q \leq 1000$): $q = 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 64, 67, 71, 73, 79, 81, 89, 97, 101, \ldots$;
- $L_3(q)$, $q \leq 200$: $q = 2, 3, 4, 5, 7, 8, 9, 13, 16, 17, 25, 49, 97, 193$;
- $U_3(q)$, $q \leq 200$: $q = 3, 4, 5, 7, 8, 11, 23, 32, 47, 128$;
- $Sz(q)$: $q = 8, 32, 128, 512, 2048, 8192$ satisfy the condition, $q = 2^{15}$ does not ($2^{15} - 1 = 7 \cdot 31 \cdot 151$);
- ${}^2G_2(q)$: $q = 27, 243, 2187$ satisfy the condition, $q = 3^9$ does not ($3^9 - 1 = 2 \cdot 13 \cdot 757$).

Whether the families 3–7 are infinite is a question of analytic number theory (simultaneous "few prime factors" conditions on values of polynomials) which we do not address. In contrast with the one-prime (EPPO) case, where Suzuki's list is finite, the answer here is necessarily given by arithmetic conditions.

1. Reduction to simple groups

Lemma 1.1. (P2) is inherited by subgroups and by quotient groups, hence by all sections. Consequently, the non-abelian composition factors of finite (P2)-groups are exactly the non-abelian simple (P2)-groups.

Proof. Subgroups: obvious. Quotients: let $N \trianglelefteq H$ and $hN \in H/N$ of order m. Since $(hN)^{|h|} = h^{|h|}N = N$, m divides $|h|$, so the prime divisors of m are among those of $|h|$ and $\omega(m) \leq \omega(|h|) \leq 2$. A composition factor H/N of G ($N \trianglelefteq H \leq G$) is a section of G. Conversely a simple (P2)-group T is a composition factor of the (P2)-group T.

So the problem is equivalent to: *classify the finite non-abelian simple groups T with (P2).*

2. Elementary lemmas

Lemma 2.1. A finite abelian group A has an element whose order has ≥ 3 prime divisors iff $\omega(|A|) \geq 3$. *Proof.* If r_1, r_2, r_3 are distinct primes dividing $|A|$, take elements a_i of order r_i (Cauchy); $a_1a_2a_3$ has order $r_1r_2r_3$. The converse is trivial.

Lemma 2.2 (number theory). For a prime power q: $\omega(q-1) \leq 1$ and $\omega(q+1) \leq 1 \iff q \in \{2,3,4,8\}$. *Proof.* If q is odd, $q-1$ and $q+1$ are even prime powers, hence powers of 2 differing by 2, so $\{q-1, q+1\} = \{2,4\}$, $q = 3$. Let $q = 2^f$. If $2^f - 1 = r^a$ with $f \geq 2$, $a \geq 2$: for a odd, $r^a + 1 = (r+1)(r^{a-1} - \cdots + 1)$ with the second factor odd and > 1, impossible for a power of 2; for a even, $r^a + 1 \equiv 2 \pmod 4$, so $2^f = 2$, contradiction. Hence $2^f - 1$ is 1 or a prime, so $f = 1$ or f is prime. If $2^f + 1 = s^b$ with $b \geq 2$: for b odd, $s^b - 1 = (s-1)(s^{b-1} + \cdots + 1)$ with odd second factor > 1, impossible; for b even, $(s^{b/2} - 1)(s^{b/2} + 1) = 2^f$ gives $s^{b/2} = 3$, $f = 3$. Otherwise $2^f + 1$ is a Fermat prime and f is a power of 2. So $f \in \{1,2,3\}$, i.e. $q \in \{2,4,8\}$; and $q = 2,3,4,8$ indeed satisfy the condition.

Lemma 2.3. Let G be a finite group, p a prime, $Z \leq Z(G)$ a p'-group, and $g = su$ with s a p'-element, u a p-element, $su = us$. Then $|gZ| = |sZ| \cdot |u|$. *Proof.* $g^k \in Z$ iff $s^k u^k \in Z$; since the decomposition of an element into commuting p'- and p-parts is unique and Z consists of p'-elements, this holds iff $u^k = 1$ and $s^k \in Z$.

Lemma 2.4. Let $T = G/Z$ with G a finite group of Lie type in characteristic p, $Z \leq Z(G)$ a p'-group.

If a p-element uZ centralises sZ (s semisimple), then u centralises s. *Proof.* $usu^{-1} = sz$ with $z \in Z$; then $u^i s u^{-i} = sz^i$, and $u^{p^k} = 1$ gives $z^{p^k} = 1$, so $z = 1$.

Lemma 2.5 (monotonicity in classical groups). Let $X(V)$ be one of $SL(V)$, $SU(V)$, $Sp(V)$, $\Omega(V)$ and $W \leq V$ a non-degenerate (any subspace for SL) subspace with a complement W' ($W' = W^\perp$ in the form cases), $W' \neq 0$. Then $X(W) \times 1_{W'} \leq X(V)$ and $(X(W) \times 1_{W'}) \cap Z(X(V)) = 1$ (scalars act as scalars on W'). Hence the simple group $X(V)/Z(X(V))$ contains a subgroup isomorphic to $X(W)$, which maps onto $X(W)/Z(X(W))$. In particular **if $X(W)/Z(X(W))$ fails (P2), so does** $X(V)/Z(X(V))$ (lift a bad element to $X(W)$, Lemma 1.1). (For Ω: the spinor norm, resp. the Dickson invariant for q even, and the determinant are multiplicative over orthogonal sums, so $\Omega(W) \times 1 \leq \Omega(V)$.)

3. Alternating groups

Proposition 3.1. A_n ($n \geq 5$) has (P2) iff $n \leq 11$. *Proof.* $(1\,2\,3\,4\,5)(6\,7\,8)(9\,10)(11\,12) \in A_{12} \leq A_n$ ($n \geq 12$) has order 30. Let $n \leq 11$ and $x \in A_n$ with $r_1 r_2 r_3 \mid |x|$ for primes $r_1 < r_2 < r_3$. If no cycle length of x is divisible by two of the r_i, the cycles cover at least $r_1 + r_2 + r_3$ points; this is ≥ 12 unless $\{r_i\} = \{2, 3, 5\}$; but then an even cycle is needed and, x being even, a second even cycle: at least $2+2+3+5 = 12$ points. If a cycle has length divisible by $r_i r_j$, it has length ≥ 6, and covering the third prime gives either $6 + 5 = 11$ (a 6-cycle and a 5-cycle: an odd permutation, excluded), or at least $10 + 3 = 13$, $6 + 7 = 13$, 15 points. Contradiction. (Also confirmed by the character tables of $A_5, \ldots, A_{19}$, §10.)

4. Sporadic groups

Proposition 4.1. The sporadic groups with (P2) are exactly M_{11}, M_{12}, M_{22}, M_{23}, M_{24}, J_1, J_2, J_3, HS, He, Ru, Suz, $O'N$. *Proof.* Inspection of the element orders (ATLAS; GAP `OrdersClassRepresentatives` of the 26 tables in the Character Table Library). The other thirteen groups Co_1, Co_2, Co_3, Fi_{22}, Fi_{23}, Fi'_{24}, HN, Ly, Th, J_4, McL, B, M all contain elements of order 30. (Script `scan_ctbllib.g`, output `scan_ctbllib.out`.)

5. Groups of Lie type: general facts and a decision procedure

Let $\mathbf{G}$ be a simple simply connected algebraic group over $\overline{\mathbb{F}}_p$, F a Steinberg endomorphism, $G = \mathbf{G}^F$, $Z = Z(G)$. Apart from $Sp_4(2)' \cong A_6$, $G_2(2)' \cong U_3(3)$, ${}^2G_2(3)' \cong L_2(8)$ and ${}^2F_4(2)'$ (treated by their tables) and the solvable groups $SL_2(2), SL_2(3), SU_3(2), Sz(2)$, every simple group of Lie type is $T = G/Z$ for such G. We use the standard facts (Carter [7, §§1.4, 3.3, 3.5], Springer–Steinberg [9], Digne–Michel [8, Ch. 3]):

- (J) every $g \in G$ is $g = su = us$ with s semisimple (p') and u unipotent (p-element);
- (T) every semisimple $s \in G$ lies in an F-stable maximal torus; the G-classes of F-stable maximal tori correspond to F-classes of the Weyl group W, and $\mathbf{T}_w^F \cong Y/(F_w - 1)Y$ where Y is the cocharacter lattice (= coroot lattice, $\mathbf{G}$ simply connected) and F_w the action of Fw on Y; equivalently $\mathbf{T}_w^F = (F_w - 1)^{-1}Y/Y \subset Y \otimes \mathbb{Q}/\mathbb{Z}$, and for $s \leftrightarrow v$, a root α satisfies $\alpha(s) = 1 \iff \langle \alpha, v \rangle \in \mathbb{Z}$;
- (S) (Steinberg) $C_{\mathbf{G}}(s)$ is connected reductive, generated by $\mathbf{T}$ and the root subgroups U_α with $\alpha \in \Phi_s := \{\alpha : \alpha(s) = 1\}$; $s \in Z(C_{\mathbf{G}}(s))$; and $C_G(s) = C_{\mathbf{G}}(s)^F$ contains a non-trivial unipotent element iff $\Phi_s \neq \emptyset$.

By Lemma 2.3 (Lemma 2.4 only confirms that working with lifts in G loses nothing): T **has (P2) iff (a)** $\omega(|sZ|) \leq 2$ **for all semisimple** $s \in G$**, and (b)** $\omega(|sZ|) \leq 1$ **for every semisimple** s **with**

$\Phi_s \neq \emptyset$. By Lemma 2.1, (a) is equivalent to $\omega(|\mathbf{T}_w^F/Z|) \le 2$ for all w. Here $|sZ|$ is the order of v modulo the coweight lattice, i.e. the lcm of the denominators of $\langle \alpha_i, v\rangle$ for the simple roots α_i.

This gives a finite decision procedure for each pair (type, q), implemented in `torus_method.g` (it enumerates $\mathbf{T}_w^F$ for one w in each F-class; for the Suzuki–Ree groups F acts on Y as $p^m\tau w$ with $\tau(\alpha^\vee) = \alpha_\rho^\vee$ or $p\,\alpha_\rho^\vee$ according as $\alpha^\vee$ is short or long, $q = p^{2m+1}$). As a control, the program also checks Steinberg's theorem that the number of F-stable maximal tori is $|G|_p^2$, i.e. $\frac{1}{|W|}\sum_w |\mathbf{T}_w^F|^{-1} = |G|_p^2/|G|$; in all 55 test cases it reproduced $|G|$ exactly. The procedure was validated against the character tables of 54 simple groups of Lie type (all verdicts agree, §10). It is used as the *main* argument only for $G_2(8)$ (for which a hand proof is also given in §8.3), and otherwise as an independent check.

6. The rank-one-type families

Proposition 6.1 ($L_2(q)$)**.** $L_2(q)$, $q \ge 4$, has (P2) iff $\omega(\frac{q-1}{d}), \omega(\frac{q+1}{d}) \le 2$, $d = (2, q-1)$. *Proof.* By Dickson's theorem [2, II.8.3–8.5, 8.27] every element of $L_2(q)$ lies in a Sylow p-subgroup (elementary abelian) or in a cyclic subgroup of order $(q-1)/d$ or $(q+1)/d$, and such cyclic subgroups exist. So the element orders are p and the divisors of $(q \pm 1)/d$.

Proposition 6.2 ($Sz(q)$)**.** $Sz(q)$ has (P2) iff $\omega(q-1), \omega(q \pm \sqrt{2q}+1) \le 2$. *Proof.* By Suzuki [3] (see also Shi [5]) the element orders of $Sz(q)$ are exactly the divisors of 4, $q-1$, $q+\sqrt{2q}+1$, $q-\sqrt{2q}+1$, and cyclic subgroups of these orders exist.

Proposition 6.3 ($^2G_2(q)$)**.** $^2G_2(q)$, $q = 3^{2m+1} \ge 27$, has (P2) iff $\omega(q-1), \omega(q+1), \omega(q \pm \sqrt{3q}+1) \le 2$. *Proof.* By Brandl–Shi [4, Lemma 4] (see also Ward [6]) the set of element orders of $^2G_2(q)$ is the set of divisors of $6, 9, q-1, \frac{q+1}{2}, q \pm \sqrt{3q}+1$; since $4 \mid q+1$, $\omega(\frac{q+1}{2}) = \omega(q+1)$.

Proposition 6.4 ($L_3(q)$ **and** $U_3(q)$)**.** Let $\varepsilon = +1$ ($L_3(q)$) or $\varepsilon = -1$ ($U_3(q)$, $q \ge 3$), $d = (3, q-\varepsilon)$. Then $L_3^\varepsilon(q)$ has (P2) iff

$$\omega\Big(\frac{q-\varepsilon}{d}\Big) \le 1, \qquad \omega\Big(\frac{q^2-1}{d}\Big) \le 2, \qquad \omega\Big(\frac{q^2+\varepsilon q+1}{d}\Big) \le 2.$$

Proof. Let $S = SL_3(q)$ acting on $V = \mathbb{F}_q^3$, resp. $S = SU_3(q)$ acting on $V = \mathbb{F}_{q^2}^3$ with a non-degenerate hermitian form; $Z = Z(S)$, $|Z| = d$, $T = S/Z$. Let $\mu_{q-\varepsilon} \le \overline{\mathbb{F}}_p^*$ be the group of $(q-\varepsilon)$-th roots of unity.

Semisimple elements. Define abelian subgroups of S: A_1 = the diagonal subgroup ($\varepsilon = -1$: diagonal w.r.t. an orthonormal basis), $|A_1| = (q-\varepsilon)^2$; A_2: for $\varepsilon = +1$, $\{x \oplus N(x)^{-1}\}$ on $\mathbb{F}_{q^2} \oplus \mathbb{F}_q$, $x \in \mathbb{F}_{q^2}^*$; for $\varepsilon = -1$, $\{x$ on e, x^{-q} on f, x^{q-1} on $w\}$ with (e,f) a hyperbolic pair and w anisotropic, $x \in \mathbb{F}_{q^2}^*$; $|A_2| = q^2-1$; A_3: the elements of determinant 1 of the "Singer" torus $\mathbb{F}_{q^3}^*$ (resp. of $\mu_{q^3+1} \le \mathbb{F}_{q^6}^*$ acting on $\mathbb{F}_{q^6} \cong \mathbb{F}_{q^2}^3$ with the trace hermitian form); $|A_3| = q^2 + \varepsilon q + 1$ (for $\varepsilon = -1$: $\det x = x^{1+q^2+q^4} = x^{q^2-q+1}$ on μ_{q^3+1}). Each A_i contains Z. Let $s \in S$ be semisimple; we show that s lies in an abelian subgroup $A \le S$ with $Z \le A$ and $|A| \in \{|A_1|, |A_2|, |A_3|\}$. For $\varepsilon = +1$, according to the factorisation of the characteristic polynomial into irreducibles over $\mathbb{F}_q$ (degrees $1+1+1$, $1+2$, 3), s is diagonalisable over $\mathbb{F}_q$, or acts as multiplication by some $x \in \mathbb{F}_{q^2}^*$ on a 2-dimensional summand $\cong \mathbb{F}_{q^2}$ and by $N(x)^{-1}$ on a complementary line, or generates $\mathbb{F}_q[s] \cong \mathbb{F}_{q^3}$; so s lies in a $GL_3(q)$-conjugate of A_1, A_2 or A_3. For $\varepsilon = -1$, decompose V into the primary components V_f of s (f irreducible over $\mathbb{F}_{q^2}$); $V_f \perp V_g$ unless $g = f^*$ (the polynomial whose roots are the λ^{-q}). A self-dual f of degree 2 is impossible (it would force $\lambda^{q+1} = 1$, i.e. $\lambda \in \mathbb{F}_{q^2}$), so there are three cases. (i) All f are self-dual of degree 1: the eigenspaces are non-degenerate and orthogonal, so s is diagonal in an orthonormal basis, i.e. lies in a conjugate of A_1. (ii) $V = V_f \oplus V_{f^*} \perp \langle w\rangle$ with $\deg f = 1$,

$f \neq f^*$: choose $e \in V_f, f' \in V_{f^*}$ with $(e,f') = 1$; then $s = \mathrm{diag}(x, x^{-q}, x^{q-1})$ in the basis e, f', w (the last entry is forced by $\det s = 1$), so s lies in a conjugate of A_2. (iii) f self-dual of degree 3: then $K = \mathbb{F}_{q^2}[s] \cong \mathbb{F}_{q^6}$, the adjoint involution of the form restricts to the automorphism $y \mapsto y^{q^3}$ of K (the only one of order 2 extending $y \mapsto y^q$ on $\mathbb{F}_{q^2}$), so $A := \{y \in K^* : y^{q^3+1} = 1,\ \det y = 1\}$ is an abelian subgroup of S containing s and Z, of order $q^2 - q + 1$. In all cases a GL_3- or GU_3-conjugate of A_i is again an abelian subgroup of S of order $|A_i|$ containing Z. Hence the semisimple elements of T are exactly the elements of the subgroups A/Z, where $A \geq Z$ runs over abelian subgroups of S of orders $|A_1|, |A_2|, |A_3|$ (the A_i themselves among them), and by Lemma 2.1 they satisfy (P2) iff $\omega((q-\varepsilon)^2/d) = \omega(q-\varepsilon) \leq 2$, $\omega((q^2-1)/d) \leq 2$, $\omega((q^2+\varepsilon q+1)/d) \leq 2$.

Elements with non-trivial unipotent part. Let $g = su$, $u \neq 1$ unipotent, $su = us$. If s had three distinct eigenvalues its centraliser in $GL_3(\overline{\mathbb{F}}_p)$ would be a torus, so s has eigenvalues $\lambda, \lambda, \lambda^{-2}$. If $\lambda^3 = 1$, then $s \in Z$ and $gZ = uZ$ has p-power order, which is harmless; so let $\lambda^3 \neq 1$. The double eigenvalue is fixed by $x \mapsto x^q$ (resp. $x \mapsto x^{-q}$), so $\lambda \in \mu_{q-\varepsilon}$. u preserves the 2-dimensional λ-eigenspace and acts trivially on the λ^{-2}-eigenline, so $|u| = p$. Moreover $s^k \in Z$ iff $\lambda^{3k} = 1$, so $|sZ| = |\lambda^3|$, and by Lemma 2.3 $|gZ| = p\,|\lambda^3|$. As λ runs over $\mu_{q-\varepsilon}$, λ^3 runs over a cyclic group of order $(q-\varepsilon)/d$; conversely, for every $\lambda \in \mu_{q-\varepsilon}$, $s = \mathrm{diag}(\lambda, \lambda, \lambda^{-2})$ (orthonormal basis if $\varepsilon = -1$) lies in S and centralises the non-trivial unipotent elements of SL (resp. SU) of its λ-eigenspace. Hence these elements have (P2) iff $\omega((q-\varepsilon)/d) \leq 1$; this also implies $\omega(q-\varepsilon) \leq 2$.

(Consistent with Buturlakin's description of the spectra of $L_3^\varepsilon(q)$ [10]; verified for all $q \leq 200$ by the procedure of §5, §10.)

7. All other groups of Lie type: reduction to $q \in \{2,3,4,8\}$

Proposition 7.1. Let T be a simple group of Lie type over $\mathbb{F}_q$ not of type $A_1, A_2, {}^2A_2, {}^2B_2, {}^2G_2$. If T has (P2) then $q \in \{2,3,4,8\}$, and T is not of type $F_4, E_6, {}^2E_6, E_7, E_8$.

Proof. We exhibit elements of orders $p(q-1)$ and $p(q+1)$; then (P2) forces $\omega(q \pm 1) \leq 1$ and Lemma 2.2 gives $q \in \{2,3,4,8\}$. Recall $SL_2(q)$ has cyclic subgroups of orders $q-1$ and $q+1$.

- $L_n(q)$, $n \geq 4$; $U_n(q)$, $n \geq 4$; $PSp_{2n}(q)$, $n \geq 2$: the natural module contains $W_1 \oplus W_2$ ($\oplus W_3$) with W_1, W_2 two-dimensional (non-degenerate in the unitary/symplectic case) and $SL(W_i) = SU(W_i) = Sp(W_i) \cong SL_2(q)$. Take $u \neq 1$ unipotent in $SL(W_1)$ and $y \in SL(W_2)$ of order $q \pm 1$, $g = u \oplus y \oplus 1$. If g^k is scalar λ, then $u^k = \lambda \cdot 1_{W_1}$ is unipotent, so $\lambda = 1$; hence $|gZ| = p\,(q \pm 1)$.
- $\Omega_{2n+1}(q)$, q odd, $n \geq 3$, and $P\Omega^\pm_{2n}(q)$, $n \geq 4$, q odd: by Lemma 2.5 they contain $\Omega_5(q) \cong PSp_4(q)$, which has such elements. $P\Omega^\pm_{2n}(q)$, q even, $n \geq 4$: contains $Sp_{2n-2}(q) \geq Sp_4(q)$ (stabiliser of a non-singular vector, [16, 4.1.7]).
- $G_2(q)$: contains $SL_3(q)$ and $SU_3(q)$ ([11], [12]); in them $s = \mathrm{diag}(\lambda, \lambda, \lambda^{-2})$ with $|\lambda| = q \mp 1$, times a commuting unipotent element of order p of SL (resp. SU) of the λ-eigenspace, has order $p(q \mp 1)$; these subgroups are contained in $G_2(q)$ faithfully.
- ${}^3D_4(q) \geq G_2(q)$ [13]; ${}^2F_4(q) \geq Sp_4(q)$, $q = 2^{2m+1}$ [14].
- $F_4(q) \geq {}^3D_4(q).3$ [15]; by the above and §8 (${}^3D_4(q)$ fails (P2) for every $q \neq 2$) and the table of $F_4(2)$ (element of order 30), $F_4(q)$ fails (P2) for all q. [Alternatively $F_4(q) \geq B_4(q)$, and $\Omega_9(q)$, $Sp_8(q)$ fail.] Finally $F_4(q)$ embeds in $E_6(q)$, ${}^2E_6(q)$ (fixed points of a graph automorphism of E_6^{sc}), and in $E_7(q)$, $E_8(q)$ (via the Levi subgroups $E_6^{sc} \leq E_7^{sc}$, $E_7^{sc} \leq E_8$; $F_4(q)$ is simple, so it meets the centre trivially).

8. The cases $q \in \{2,3,4,8\}$

8.1 Classical groups

By Lemma 2.5, once $X_m(q)$ fails, all $X_n(q)$, $n \geq m$, fail. The base cases (GAP character tables unless stated):

family	(P2) holds	first failure (witness order)
$L_n(2)$, $n \geq 4$	$L_4(2) \cong A_8$, $L_5(2)$	$L_6(2)$ (30)
$L_n(3)$, $n \geq 4$	$L_4(3)$	$L_5(3)$ (78)
$L_n(4)$, $n \geq 4$	—	$L_4(4)$ (30)
$L_n(8)$, $n \geq 4$	—	$L_4(8)$ ($126 = 2 \cdot 3^2 \cdot 7$; explicit matrix, `boundary_check.g`)
$U_n(2)$, $n \geq 4$	$U_4(2)$, $U_5(2)$, $U_6(2)$	$U_7(2)$ (30)
$U_n(3)$, $n \geq 4$	$U_4(3)$	$U_5(3)$ (42)
$U_n(4)$, $n \geq 4$	—	$U_4(4)$ (30)
$U_n(8)$, $n \geq 4$	—	$U_4(8)$ ($455 = 5 \cdot 7 \cdot 13$; explicit element of $SU_4(8)$, centre trivial)
$PSp_{2n}(2)$, $n \geq 3$	$PSp_6(2)$	$PSp_8(2)$ (30)
$PSp_{2n}(3)$, $n \geq 2$	$PSp_4(3) \cong U_4(2)$	$PSp_6(3)$ (30)
$PSp_{2n}(4)$, $n \geq 2$	$PSp_4(4)$	$PSp_6(4)$ (30)
$PSp_{2n}(8)$, $n \geq 2$	$PSp_4(8)$	$PSp_6(8)$ (455; explicit element of $Sp_6(8)$)
$\Omega_{2n+1}(3)$, $n \geq 3$	$\Omega_7(3)$	$\Omega_9(3)$ (30)
$P\Omega^+_{2n}(2)$, $n \geq 4$	$P\Omega^+_8(2)$	$P\Omega^+_{10}(2)$ (30)
$P\Omega^-_{2n}(2)$, $n \geq 4$	—	$P\Omega^-_8(2)$ (30)
$P\Omega^+_{2n}(3)$, $n \geq 4$	$P\Omega^+_8(3)$	$P\Omega^+_{10}(3)$ (30)
$P\Omega^-_{2n}(3)$, $n \geq 4$	—	$P\Omega^-_8(3)$ (30)
$P\Omega^\pm_{2n}(4), P\Omega^\pm_{2n}(8)$, $n \geq 4$	—	contain $Sp_6(q)$, which fails

Monotonicity used: $L_n \supseteq SL_{n-1}$, $U_n \supseteq SU_{n-1}$, $PSp_{2n} \supseteq Sp_{2n-2}$; $\Omega_{2n+1}(3)$ ($n \geq 5$) and $P\Omega^\pm_{2n}(3)$ ($n \geq 5$) contain $\Omega_9(3)$; $P\Omega^\pm_{2n}(2) \supseteq Sp_{2n-2}(2) \supseteq Sp_8(2)$ for $n \geq 5$; $P\Omega^-_{2n}(q) \supseteq \Omega^-_8(q)$. ($\Omega_{2n+1}(q)$ for q even is $PSp_{2n}(q)$; orthogonal groups of dimension ≤ 6 are L_2, L_4, U_4, PSp_4.) In particular, $PSp_4(q)$ has (P2) exactly for $q \in \{3,4,8\}$.

8.2 Exceptional groups

- $G_2(q)$, $q \in \{3,4,8\}$: all three have (P2) ($G_2(3), G_2(4)$: tables; $G_2(8)$: §8.3). ($G_2(2)' \cong U_3(3)$.)
- ${}^3D_4(q)$: ${}^3D_4(2)$ has (P2) (table); ${}^3D_4(3)$ (42), ${}^3D_4(4)$ (42) fail (tables); ${}^3D_4(8)$ fails: it contains $SL_2(512) \times SL_2(8)$ [13], hence an element of order $513 \cdot 7 = 3^3 \cdot 7 \cdot 19$ (also found by the procedure of §5). Other q: Proposition 7.1.
- ${}^2F_4(q)$: ${}^2F_4(2)'$ and ${}^2F_4(8)$ have (P2) (tables; for ${}^2F_4(8)$ also §5: the eleven maximal tori have orders 25, 35, 37, 49, 57, 63, 65, 81, 91, 109, 169, and every semisimple $s \neq 1$ with $\Phi_s \neq \emptyset$ has order dividing 3, 7, 9, 5 or 13). For $q = 2^{2m+1} \geq 32$: Proposition 7.1.
- F_4, E_6, ${}^2E_6$, E_7, E_8: fail (Proposition 7.1).

8.3 Hand proof that $G_2(8)$ has (P2)

G_2 is simply connected and adjoint. (a) The F-classes of $W = D_{12}$ give six classes of maximal tori, with $|\mathbf{T}_w^F| = \chi_w(q)$, χ_w the characteristic polynomial of w: $(q-1)^2, (q+1)^2, q^2-1$ (two classes), q^2+q+1, q^2-q+1. For $q = 8$: $49, 81, 63 = 3^2 \cdot 7, 73, 57 = 3 \cdot 19$, all with $\omega \leq 2$. (b) With a short and b long simple roots: if $\alpha_1, \ldots, \alpha_k \in \Phi_s$ and $\beta \in \Phi \cap \sum \mathbb{Z}\alpha_i$ then $\beta(s) = 1$, so

$\Phi_s = \Phi \cap \mathbb{Z}\Phi_s$; in particular Φ_s is a root subsystem. The root subsystems of G_2 are, up to W, $\emptyset$, $A_1, \tilde{A}_1, A_1\tilde{A}_1 = \{\pm b, \pm(2a+b)\}, A_2 = \{\pm b, \pm(3a+b), \pm(3a+2b)\}, \tilde{A}_2$ (short roots) and G_2; $\tilde{A}_2$ does not occur since the short roots span the whole root lattice, and $\Phi_s = G_2$ means $s \in Z(\mathbf{G}) = 1$. Writing $t \in \mathbf{T}$ as $(a(t), b(t)) = (x, y)$ (recall $X(\mathbf{T}) = \mathbb{Z}a \oplus \mathbb{Z}b$): for $A_1\tilde{A}_1$, $Z(C_{\mathbf{G}}(s)) = \{y = 1,\ x^2y = 1\} = \{y = 1, x^2 = 1\}$ is trivial in characteristic 2; for A_2 it is $\{y = 1,\ x^3y = 1\}$, of order 3; for A_1 or $\tilde{A}_1$ it is the kernel of a primitive character, hence a one-dimensional torus $\mathbf{S}$, F-stable because $C_{\mathbf{G}}(s)$ is, and F acts on $X(\mathbf{S}) \cong \mathbb{Z}$ as $\pm q$, so $|\mathbf{S}^F| = q \mp 1$. So every semisimple $s \neq 1$ with $\Phi_s \neq \emptyset$ has order dividing 3, 7 or 9, a prime power. By §5, $G_2(8)$ has (P2).

9. Conclusion of the proof

Every non-abelian simple group is alternating, sporadic, or of Lie type. §§3–4 settle the first two classes. For Lie type: the families $L_2, L_3, U_3, Sz, {}^2G_2$ are settled by §6 (for every q); by Proposition 7.1 every other group of Lie type with (P2) is defined over $q \in \{2, 3, 4, 8\}$ and is not of type $F_4, E_6, {}^2E_6, E_7, E_8$; §8 lists exactly which of these have (P2): item 8 of the Main Theorem together with $L_4(2) \cong A_8$ and $PSp_4(2)' \cong A_6$. With Lemma 1.1 this proves the Main Theorem.

10. Computer verification (GAP 4.14, CTblLib, all scripts in this folder)

- `scan_ctbllib.g` → `scan_ctbllib.out`: all 154 non-abelian simple groups with a character table in CTblLib (all 26 sporadic groups, A_5–A_{19}, many groups of Lie type including $S_4(8)$ and ${}^2F_4(8)$). 85 have (P2), 69 do not (with a witness element order).
- `compare_theorem_with_tables.py` → `compare_theorem_with_tables.out`: the Main Theorem predicts the verdict of every one of these 154 tables correctly (no mismatches).
- `torus_method.g`: the decision procedure of §5.
- `validate_torus_method.g` → `validate_torus_method.out`: 55 groups of Lie type; for the 54 with a table the procedure agrees with the table, and Steinberg's count reproduces $|G|$ in all cases.
- `families_check.g` → `families_check.out`: the closed-form criteria of §6 agree with the procedure for $L_2(q)$, $q \leq 1000$ (191 values), $L_3(q)$ and $U_3(q)$, $q \leq 200$, $Sz(q)$, $q \leq 2^{11}$, ${}^2G_2(q)$, $q \leq 3^7$: 0 mismatches. `families_check2.g` → `families_check2.out` extends the $L_2(q)$ check to $1000 < q \leq 2000$ (140 values, 0 mismatches). `families_check3.g` → `families_check3.out` extends the $L_3(q)$, $U_3(q)$ check to $200 < q \leq 400$ (37 values, 0 mismatches). `direct_L3U3_check.g` → `direct_L3U3_check.out` checks the criterion of Proposition 6.4 directly on the conjugacy classes of the actual groups $L_3(q)$, $q \leq 13$, and $U_3(q)$, $3 \leq q \leq 13$ (17 groups, 0 mismatches).
- `mixed_orders.g` → `mixed_orders.out`: for $G_2(8)$, ${}^2F_4(8)$, $PSp_4(8)$ the torus orders and the orders of the semisimple $s \neq 1$ with $\Phi_s \neq \emptyset$ (resp. $\{3, 7, 9\}$, $\{3, 5, 7, 9, 13\}$, $\{3, 7, 9\}$), confirming §8.2–8.3.
- `extra_torus_check.g` → `extra_torus_check.out`: the procedure confirms that $U_7(2)$, $PSp_6(4)$, $\Omega_9(3)$, $P\Omega^{\pm}_{10}(2)$, $P\Omega^{\pm}_{10}(3)$ fail (P2).
- `direct_classes_check.g` → `direct_classes_check.out` (no character tables used): conjugacy classes of the actual groups $L_4(3)$, $L_5(2)$, $U_4(3)$, $U_5(2)$, $PSp_4(4)$, $PSp_6(2)$, $PSp_4(8)$, $P\Omega^+_8(2)$ (permutation action on projective points) give the same element orders, all with $\omega \leq 2$. `direct_classes_check2.g` → `direct_classes_check2.out` does the same for $\Omega_7(3)$ and $P\Omega^+_8(3)$ (element orders 1–10, 12, 13, 14, 15, 18, 20 in both). `direct_classes_check3.g` → `direct_classes_check3.out`: $U_6(2)$ (action of $SU_6(2)$ on the 693 isotropic points; element orders 1–12, 15, 18).
- `g2_8_sample.g` → `g2_8_sample.out`: an independent construction of $G_2(8)$ from the Chevalley lattice of the 7-dimensional module (order 4329310519296 confirmed); 3000 random elements have orders in $\{3, 4, 6, 7, 8, 9, 12, 14, 18, 19, 21, 57, 63, 73\}$, none with three prime di-

visors (evidence only; the proof is §8.3).

- `direct_L2_check.g` → `direct_L2_check.out`: the criterion of Proposition 6.1 agrees with the conjugacy classes of $PSL(2,q)$ for all 42 prime powers $4 \le q \le 128$.
- `boundary_check.g` → `boundary_check.out`: $L_4(8)$, $U_4(8)$, $PSp_6(8)$, ${}^3D_4(4)$, ${}^3D_4(8)$ fail, $G_2(8)$ has (P2) (procedure); explicit elements of orders 126 in $SL_4(8)$, 30 in $SL_4(4)$, 455 in $SU_4(8)$ and $Sp_6(8)$, 30 in $Sp_6(4)$ (centres trivial).

All scripts are plain GAP 4.14 files (`gap -q <script>.g`); `torus_method.g` is read by the other scripts.

11. Status of the argument

- **Proved by hand (modulo standard theory):** reduction (§1), alternating groups (§3), the families $L_3(q)$, $U_3(q)$ (§6.4), the reduction to $q \in \{2,3,4,8\}$ and the bounds on rank (§7, §8.1 monotonicity), $G_2(8)$ (§8.3).
- **Relies on cited classical results:** spectra of $L_2(q)$ (Dickson), $Sz(q)$ (Suzuki), ${}^2G_2(q)$ (Ward; Brandl–Shi); the subgroup containments $SL_3(q), SU_3(q) \le G_2(q), G_2(q)$, $SL_2(q^3) \circ SL_2(q) \le {}^3D_4(q)$, $Sp_4(q) \le {}^2F_4(q)$, ${}^3D_4(q).3 \le F_4(q)$, $F_4(q) \le E_6(q), {}^2E_6(q), E_7(q), E_8(q)$; the general theory of maximal tori and semisimple centralisers (Steinberg, Springer–Steinberg, Carter).
- **Relies on computation with library character tables (ATLAS/CTblLib):** the sporadic groups and the finitely many base cases in §8 ($L_4(3)$, $L_5(2)$, $L_6(2)$, $L_5(3)$, $L_4(4)$, $U_4(2)$, $U_4(3)$, $U_4(4)$, $U_5(2)$, $U_5(3)$, $U_6(2)$, $U_7(2)$, $PSp_4(4)$, $PSp_4(8)$, $PSp_6(2)$, $PSp_6(3)$, $PSp_6(4)$, $PSp_8(2)$, $\Omega_7(3)$, $\Omega_9(3)$, $P\Omega_8^{\pm}(2)$, $P\Omega_8^{\pm}(3)$, $P\Omega_{10}^{\pm}(2)$, $P\Omega_{10}^{+}(3)$, $G_2(3)$, $G_2(4)$, ${}^3D_4(2)$, ${}^3D_4(3)$, ${}^3D_4(4)$, ${}^2F_4(2)'$, ${}^2F_4(8)$, $F_4(2)$). Each of these verdicts except the one for ${}^2F_4(2)'$ (not of the form $\mathbf{G}^F/Z$) is independently reproduced by the procedure of §5 (`validate_torus_method.out`, `extra_torus_check.out`, `boundary_check.out`).
- An independent referee pass (independent code and GAP scripts: character tables, direct conjugacy classes of $PSL(3,q)$ for 22 values $q \le 64$ and $PSU(3,q)$ for $q \le 13$, an independent construction of $G_2(8)$) found no mathematical error; its minor wording remarks are incorporated.
- Nothing in the classification is conjectural. The only open aspect is number-theoretic: whether each of the families 3–7 contains infinitely many groups.
- Literature search (arXiv API, web search, October 2026) found no previous classification of simple groups with (P2); Problem 20.87 is still listed as open in arXiv:1401.0300v48.

Appendix A. Maximal element orders of the groups in items 2 and 8 (from CTblLib; `spectra_list.g` → `spectra_list.out`)

group	maximal element orders $\mu(T)$
M11	5,6,8,11
M12	6,8,10,11
M22	5,6,7,8,11
M23	6,8,11,14,15,23
M24	8,10,11,12,14,15,21,23
J1	6,7,10,11,15,19
J2	7,8,10,12,15
J3	8,9,10,12,15,17,19
HS	7,8,11,12,15,20
He	8,10,12,15,17,21,28
Ru	14,15,16,20,24,26,29
Suz	11,13,14,15,18,20,21,24
ON	11,12,15,16,19,20,28,31

group	maximal element orders $\mu(T)$
L4(3)	8,9,12,13,20
L5(2)	8,12,14,15,21,31
U4(2)	5,9,12
U4(3)	5,7,8,9,12
U5(2)	8,11,12,15,18
U6(2)	7,8,10,11,12,15,18
S4(4)	4,6,10,15,17
S4(8)	4,14,18,63,65
S6(2)	7,8,9,10,12,15
O7(3)	8,12,13,14,15,18,20
O8+(2)	7,8,9,10,12,15
O8+(3)	8,12,13,14,15,18,20
G2(3)	7,8,9,12,13
G2(4)	8,10,12,13,15,21
3D4(2)	8,12,13,18,21,28
2F4(2)′	10,12,13,16
2F4(8)	12,16,18,20,28,35,37,52,57,63,65,91,109
G2(8)	(no library table; (P2) proved in §8.3)

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